Cold Fusion in a Metal: A Restoring Seat in an Octahedral Cage

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Abstract

Cold fusion, in this theory, is the slow charged entrance of a like pair at room temperature. Its host along one axis is a cage of like ions. A regular octahedron of positive ions holds one further positive ion at its center and pushes an electron off that center, so the electrons of the metal are not a screen. Two protons on an axis of the same octahedron, of cage radius 2ℓ, have a separation-increasing component that is strictly negative at separation 2ℓ, for every ion charge at least one. Just outside the first charge node the component tends to $+\infty$: the partner's repulsion diverges, and the six ions stay off the node. Between the node and separation 2ℓ the component falls strictly, and it vanishes at exactly one separation. Toward the node the component is outward, and toward 2ℓ it is inward, so both sides drive the pair to that separation. The cage does not carry the pair through the node. The seat does not set the ratio of the two lengths, and it does not derive a rate, an excess of heat, or a choice among experiments.

1 Reports of an entrance in a metal

Fleischmann and Pons placed deuterium in palladium by the electrolysis of heavy water and reported an excess of heat, together with neutrons and tritium [3]. Jones and others, in the same year, reported a low yield of neutrons from electrolytic cells [4].

Yamaguchi and Nishioka held a deuterium-loaded palladium plate in vacuum behind an oxide barrier, and reported a burst of neutrons, a release of gas, and an excess of heat [5]. Arata and Zhang placed fine particles of palladium inside a closed palladium cathode, electrolyzed heavy water, and reported an excess of heat and helium-4 [6]. Mizuno and others reported an isotopic anomaly in the material deposited on a palladium cathode [7]. Iwamura, Sakano, and Itoh permeated deuterium gas at 343 K and one atmosphere through a palladium multilayer that contained calcium oxide. They reported that cesium on the surface decreased and praseodymium appeared, and that strontium decreased and molybdenum appeared [8].

The New Hydrogen Energy project of the New Energy and Industrial Technology Development Organization, closed in 1998, did not reproduce an excess of heat in the open electrolytic cells it ran [9]. In a project of the same organization from 2015 to 2017, Kitamura, Takahashi, Iwamura, Kasagi, and others passed deuterium gas, or hydrogen gas, through a zirconia-supported nanocomposite of palladium and nickel, and through one of copper and nickel. They reported an excess of heat at room temperature and near 200–300 °C [14]. A divided sample, run in a second laboratory, gave the same report [15].

Miles and others reported that helium-4 in the electrolysis gas rose with the excess power of a palladium cathode in heavy water. Controls run in light water gave no helium-4 and no excess of power [10]. McKubre and others reported an excess of power once the deuterium loading of palladium was high [11]. Focardi, Habel, and Piantelli loaded nickel with hydrogen gas and reported an excess of heat [12].

A deuteron beam on palladium, from 2.5 to 10 keV, is a different entrance. Yuki, Kasagi, and others reported a proton yield of the deuteron–deuteron reaction above the extrapolation from higher energy [13]. The kinetic energy of the beam lies far above the kinetic energy of room temperature. The seat computed below is the seat of the room-temperature pair.

The hosts that return are palladium and nickel. Section 2 takes the octahedral hole of their face-centered cubic arrangement as the cage of a like pair. The seat does not choose among these reports, and it does not derive a rate or an excess of heat.

2 The cage

Cold fusion is the slow charged entrance fixed by the nuclear companion [1]. It is the entrance of a like pair whose kinetic energy is that of room temperature. Below the rapidity threshold the nuclear term opposes the Coulomb term, and next to the threshold it exceeds every finite ratio of the two prefactors, so long as the electromagnetic factor stays positive. A free infall meets the threshold before the node. The host of the holding, along one axis, is the electromagnetic channel already fixed for charges [2]. No second coupling is introduced.

On the equal-scale locus the force that increases the separation of two charges is

$$F(r) = \frac{kq_1q_2}{\gamma_s(r)\, r^2}, \qquad k = \frac{\alpha\hbar c}{e^2}, \tag{1}$$

with $\gamma_s(x) = \cosh x \cos x - \sinh x \sin x$ and $x = \ell/(2r)$. Like charges repel where $\gamma_s > 0$ and attract where $\gamma_s < 0$. The first node x_1 is the unique root of $\tan x = \coth x$ in $(\pi/4, \pi/2)$, at $r_1/\ell = 0.53330$. The outer well is $r > r_1$, where $\gamma_s > 0$.

A regular octahedron of positive ions holds a further positive ion at its center and pushes an electron off that center. The force at the center vanishes by the symmetry, and every displacement of the positive ion is restored, while every displacement of an electron is pushed further. Under the Coulomb law both rates are zero. That zero is Earnshaw's theorem, and the signed factor removes it. A metal in the outer well is a held arrangement of ions, and the electrons have no seat among them. Screening is not introduced. Each ion remains a point charge.

The nearest positive ions about an octahedral hole of the face-centered cubic arrangement form this octahedron. Palladium and nickel are of that arrangement. The name records the geometry of the hole. The reports of Section 1 asked whether a nuclear entrance occurred there. The seat below does not decide what that arrangement produced.

3 The axial component

The centered ion is one charge, restored in every direction. Two protons on an axis are a pair, and the component computed here is the component along that axis. Place ions of charge Ze at $(\pm a, 0, 0)$, $(0, \pm a, 0)$, and $(0, 0, \pm a)$, and protons of charge $+e$ at $(\pm t, 0, 0)$, with $0 < t < a$. Every ion–proton separation and the proton–proton separation are taken off every node. The component that increases the separation is

$$F_\parallel(t) = -ZF(a-t) + ZF(a+t) + 4Z\frac{t}{R}F(R) + F(2t), \tag{2}$$

where $R = \sqrt{a^2 + t^2}$ and F is (1) for two unit charges. The first term is the nearer axial ion. On the outer well it is a repulsion, and the direction from that ion toward the proton points inward, so the term decreases the separation. The second term is the farther axial ion, and it increases the separation. The third is the four equatorial ions: each repulsion has an outward axial projection t/R. The last term is the partner. On the outer well, $F_\parallel$ is a contest among repulsions.

A deuteron carries the same charge $+e$, so (2) is the axial component of a deuteron pair in the same cage. The coordination saddles of a metallic bond, at eight and at twelve neighbors, are saddles of a reduced electron content. They are not this interstitial seat.

4 Outward outside the node, inward at twice the length

Take the cage radius to be $a = 2\ell$, and the half-separation to be $t = \ell$. The proton separation is then 2ℓ. The nearer ion stands at ℓ, the farther at 3ℓ, each equatorial ion at $\sqrt{5}\,\ell$, and the partner at 2ℓ. Each of these distances lies in the outer well. For every ion charge $Z \geq 1$,

$$F_\parallel(\ell) < 0. \tag{3}$$

The nearer ion outweighs the farther ion, the four equatorial ions, and the partner together. Its phase is the largest, so $1/\gamma_s$ amplifies it more than the others. The cage contribution at this separation is itself inward, and a larger Z adds to that inward side. The partner does not depend on Z.

As the separation decreases to the first charge node, $2t \downarrow r_1$, the partner's phase increases to x_1 and $F(2t)$ tends to $+\infty$. The six ions remain off the node: the nearest approaches $2\ell - r_1/2$, still outside r_1,

and the other five are farther. Their contribution tends to a finite value, so

$$F_{\parallel}(t) \to +\infty \qquad \text{as} \qquad 2t \downarrow r_1. \tag{4}$$

The component is outward in a punctured right neighborhood of the node and inward at separation 2ℓ. It is continuous on the intervening separations, all of which stay in the outer well, and on those separations it falls strictly. The zero is therefore unique:

$$r_1 < s < 2\ell, \qquad F_{\parallel}(s/2) = 0. \tag{5}$$

On the side toward the node the component is outward, and on the side toward separation 2ℓ it is inward. Both sides drive the pair to that separation. The cage does not carry the pair through the node.

The divergence that stops the approach sits at the finite radius r_1, where γ_s vanishes. Under the Coulomb law the partner term diverges only at coincidence, and the three-direction sum on a single charge is zero, so the cage does not hold one proton at its center. The signed factor holds one positive ion at the center, and it stops a pair at a finite node.

5 What the seat leaves open

The seat is that one separation, and it keeps the pair in $(r_1, 2\ell)$. The relative rapidity of a free infall is the rapidity of a separation that is falling toward the node, and the seat keeps the separation off that fall. Room temperature remains the class of the nuclear companion: a kinetic energy at most 10^{-10} of the rest energy. The nuclear comparison for the pair remains that entrance. The electromagnetic factor is positive at the nuclear node, and the rapidity is below the threshold.

The seat is a statement in the electromagnetic length. At the reduced Compton wavelength of the electron, the cage radius 2ℓ lies strictly between 772.2 fm and 772.4 fm. The octahedral hole of palladium, measured from its centre to a nearest ion, is about 2 Å, larger than this radius by a factor of about 250. The balance proved here is the one at cage radius 2ℓ, in units of that length. It is not the crystallographic size of the hole. The nuclear entrance is a statement in the nuclear length. The cage does not set their ratio. If the charge node lies outside the nuclear node, the balance (5) stands outside an infinite electromagnetic wall, and the classical path on the axis does not reach the nuclear skin. The seat does not derive a rate, an excess of heat, or a choice among the experiments that have asked whether cold fusion occurs in a metal.

References

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